① 积分换坐标系.

Zylinderkoordinaten

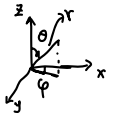
$$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}r\cos\varphi\\r\sin\varphi\\z\end{pmatrix}$$

$d(x,y,z)=r\,d(r,\varphi,z)$

Kugelkoordinaten.

$$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}r\sin\theta\cos\varphi\\r\sin\theta\sin\varphi\\r\cos\theta\end{pmatrix}$$

$d(x,y,z)=r^2\sin\theta\,d(r,\varphi,\theta)$



$e^{it}=\cos t+i\sin t$

$\cos(t)=\frac{1}{2}(e^{it}+e^{-it})$

$\sin(t)=\frac{1}{2i}(e^{it}-e^{-it})$

$\sin 2\alpha=2\sin\alpha\cos\alpha$

$\cos 2\alpha=\cos^2\alpha-\sin^2\alpha$
$=1-2\sin^2\alpha$
$=2\cos^2\alpha-1$

$\sinh x=\frac{1}{2}(e^x-e^{-x})$

$\cosh x=\pm(e^x+e^{-x})$

$\int\frac{dx}{1+x^2}=\arctan x+C$

$\int\frac{dx}{\sqrt{1-x^2}}=\arcsin x+C$

$\int\frac{dx}{\cos^2 x}=\tan x+C$

$\int\frac{dx}{\sin^2 x}=-\cot x+C$

$\int a^x\,dx=\frac{a^x}{\ln a}+C$

$\int\sinh x\,dx=\cosh x+C$

$\int\cosh x\,dx=\sinh x+C$

$\int\frac{dx}{a^2+x^2}=\frac{1}{a}\arctan\frac{x}{a}+C$

$\int\frac{dx}{a^2-x^2}=\frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right|+C$

$\int\frac{dx}{\sqrt{a^2-x^2}}=\arcsin\frac{x}{a}+C$

$\int\frac{dx}{\sqrt{x^2+a^2}}=\ln(x+\sqrt{x^2+a^2})+C$

$\int\frac{dx}{\sqrt{x^2-a^2}}=\ln|x+\sqrt{x^2-a^2}|+C$

② 三重积分的一些计算.

Schwerpunkt.

$$\begin{pmatrix}x\\y\\z\end{pmatrix}_S=\frac{\int_D\begin{pmatrix}x\\y\\z\end{pmatrix}\rho(x,y,z)\,d(x,y,z)}{\int_D\rho(x,y,z)\,d(x,y,z)}$$

Trägerheitsmoment.

$\Theta=\int_D\rho(x,y,z)\,r^2(x,y,z)\,d(x,y,z)$

z.B. bzgl. x , $r^2=y^2+z^2$

曲线，曲面积分.

$$\int_C V\,d\vec{s}=\int_a^b V(c(t))\cdot c'(t)\,dt$$

$$\int_C\langle V,n\rangle\,dx=\int_a^b\langle V(c(t)),n(c(t))\rangle\,|n(c(t))|\,d(t)$$

$$c(t)=\begin{pmatrix}f_1(t)\\f_2(t)\end{pmatrix}\quad c'(t)=\begin{pmatrix}f_1'(t)\\f_2'(t)\end{pmatrix}\quad n(c(t))=\begin{pmatrix}f_2'(t)\\-f_1'(t)\end{pmatrix}\Big/|n(c(t))|$$

曲面积分. Flächeninhalt.

① 非柱体.

投影在 x_0y 平面. 积分区域为这个投影

$$\int_F f\,d\sigma=\int_D f\sqrt{1+z_x^2+z_y^2}\,d(x,y)$$

② 柱体.

$$P(\varphi,z)=\begin{pmatrix}R\cos\varphi\\R\sin\varphi\\z\end{pmatrix}\quad\left\|\frac{\partial P}{\partial\varphi}\times\frac{\partial P}{\partial z}\right\|=R$$

$$\int_F f\,d\sigma=\int_D f(c(\varphi,z))\cdot R\,d(\varphi,z)$$

$$\int_\partial\langle f,n\rangle\,d\sigma=\int_D\langle f(c(u)),n(c(u))\rangle\left\|\frac{\partial P}{\partial u_1}\times\frac{\partial P}{\partial u_2}\right\|d(u_1,u_2)$$

$$n(c(u))=\frac{\frac{\partial P}{\partial u_1}\times\frac{\partial P}{\partial u_2}}{\left\|\frac{\partial P}{\partial u_1}\times\frac{\partial P}{\partial u_2}\right\|}$$

积分定理.

$$\operatorname{grad}f=\nabla f=\begin{pmatrix}\frac{\delta f}{\delta x}\\\frac{\delta f}{\delta y}\end{pmatrix}\qquad \operatorname{div}f=\nabla\cdot f=\frac{\delta f_1}{\delta x}+\frac{\delta f_2}{\delta y}\qquad \operatorname{rot}f=\nabla\times f=\frac{\delta f_2}{\delta x}-\frac{\delta f_1}{\delta y}\ (R_2)\longrightarrow\begin{pmatrix}\frac{\delta f_3}{\delta y}-\frac{\delta f_2}{\delta z}\\\frac{\delta f_1}{\delta z}-\frac{\delta f_3}{\delta x}\\\frac{\delta f_2}{\delta x}-\frac{\delta f_1}{\delta y}\end{pmatrix}(R_3)$$

$\Delta f=\nabla\cdot(\nabla f)=\frac{\delta^2 f}{\delta x^2}+\frac{\delta^2 f}{\delta y^2}+\frac{\delta^2 f}{\delta z^2}$ $\quad\Delta f=0\to$ harmonische Funktion.

Satz von Gauß.

$$\oint_F\langle f,n\rangle\,d\sigma=\int_V\operatorname{div}f\,d(x,y,z)$$

Satz von Green.

$$\oint_C\langle f(x),dx\rangle=\int_D\operatorname{rot}f(x)\,dx$$

Satz von Stocks.

$$\int_F\langle\operatorname{rot}f,n\rangle\,d\sigma=\oint_C\langle f,dx\rangle\ \overset{R^3}{\Longrightarrow}\ \int_F\langle\operatorname{rot}f,n\rangle\,d\sigma=\oint_{\delta F}\langle f(x,y,z),d(x,y,z)\rangle.$$

傅立叶级数.

$$S_n(t)=\frac{a_0}{2}+\sum_{k=1}^{n}\left(a_k\cos(k\omega t)+b_k\sin(k\omega t)\right)$$

$a_k=\frac{2}{T}\int_0^T f(t)\cos(k\omega t)\,dt$ $\quad b_k=\frac{2}{T}\int_0^T f(t)\sin(k\omega t)\,dt$. $\quad\omega=\frac{2\pi}{T}$.

gerade $\Rightarrow b_k=0$, ungerade $a_k=0$ $\quad C_k=\frac{1}{T}\int_0^T e^{-ik\omega t}f(t)\,dt$

$S_n(t)=\sum_{k=-n}^{n}C_k e^{ik\omega t}$, $C_0=\frac{a_0}{2}$, $C_k=\frac{1}{2}(a_k-ib_k)$ $C_{-k}=\frac{1}{2}(a_k+ib_k)$ $k\geq 1$

Parseval'sche Gleichung.

$$\frac{a_0^2}{2}+\sum_{k=1}^{\infty}(|a_k|^2+|b_k|^2)=2\sum_{k\in\mathbb{Z}}|C_k|^2=\frac{2}{T}\int_0^T|f(t)|^2\,dt.$$

Bessel'sche Ungleichung.

$$\frac{a_0^2}{2}+\sum_{k=1}^{\infty}(|a_k|^2+|b_k|^2)=2\sum_{k=-n}^{n}|C_k|^2\leq\frac{2}{T}\int_0^T|f(t)|^2\,dt.$$

傅立叶变换.

$$\hat{f}(w)=\int_{-\infty}^{+\infty}f(t)e^{-iwt}\,dt\qquad \overset{\vee}{f}(w)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}f(t)e^{iwt}\,dt.$$

$\mathcal{F}(f(t))=\hat{f}(w)\Rightarrow\mathcal{F}(\hat{f}(t))=2\pi f(-w)$

Plancherel Gleichung.

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}|\hat{f}(w)|^2\,dw=\int_{-\infty}^{\infty}|f(x)|^2\,dt.$$

absolut integrierbar $\int_{-\infty}^{\infty}|f(t)|\,dt<\infty$

$\mathcal{F}(af+pg)=a\mathcal{F}(f)+p\mathcal{F}(g)$

$g(t)=f(t+th)\Rightarrow\mathcal{F}(g(t))=e^{ihw}\mathcal{F}(f(t))$

$\mathcal{F}(e^{-ikt}\cdot f(t))=\hat{f}(w+h)$

$g(t)=f(at)\Rightarrow\hat{g}(w)=\frac{1}{a}\hat{f}(\frac{w}{a})$

$(\hat{f}')(w)=iw\hat{f}(w)$

$g(t)=tf(t)\Rightarrow(\hat{f})'(w)=-i\hat{g}(w)$

$$f*g:=\int_{-\infty}^{\infty}f(s)\,g(t-s)\,ds$$

$\mathcal{F}(f*g)=\hat{f}\cdot\hat{g}$

拉普拉斯变换.

$$\mathcal{L}f_{(s)}=\int_0^\infty f(t)e^{-st}\,dt$$

$\mathcal{L}(af+pg)=a\mathcal{L}f+p\mathcal{L}g$

$g(t)=f(at)\Rightarrow\mathcal{L}g=\frac{1}{a}\mathcal{L}f(\frac{s}{a})$

$g(t)=e^{at}f(t)\Rightarrow\mathcal{L}g=\mathcal{L}f(s+a)$

$g(t)=f(t+a)\Rightarrow\mathcal{L}g=e^{as}\mathcal{L}f$

$\mathcal{L}(f')=s\mathcal{L}(f)-f(0)$

$g(t)=tf(t)$

$\Rightarrow\mathcal{L}(g)=-1(\mathcal{L}(f))'$

微分方程.

$y_p(t)=\Phi(t)\,C_p(t)$ $\quad C_p(t)=\int_0^t\Phi^{-1}(s)\,b(s)\,ds$

$y(t)=\Phi(t)\cdot(C+C_p(t))$

这里, $\lambda_{1,2}=a\pm ib$. 任选一个入解出 v.

$$\Phi(t)=\left\{\operatorname{Re}\{e^{\lambda t}v\},\ \operatorname{Im}\{e^{\lambda t}v\}\right\}$$

$y'=Ay$, $y(0)=y_0\Rightarrow y(t)=\exp(At)\cdot y_0$

$y'=Ay+b(t)$, $y(0)=y_0\Rightarrow y(t)=\exp(At)y_0+\int_0^t\exp(A(t-s))b(s)\,ds$

$\exp(At)=V\exp(Dt)V^{-1}$

exakt DGL

$f(t,y)y'+g(t,y)=0$

$g=\frac{\delta F}{\delta t}$ $\quad f=\frac{\delta F}{\delta y}$ $\Rightarrow f_t=g_y$

↳ 把 F 解出来, 让 $F(t,y)=C$ 解 y.

复分析.

复可导. $f(z)=f(x+iy)=u(x,y)+iv(x,y)$

$u_x=v_y$

$u_y=-v_x$. $\quad f'(x+iy)=v_x+iv_x$

$\sum_{k=0}^{\infty} a_k(z-z_0)^k$

$r=\frac{1}{\lim_{k\to\infty}\sqrt[k]{|a_k|}}$

奇点类型.

$\lim_{z\to z_0} f(z)=a$ 一个定值. Hebbare Singularität.

$\lim_{z\to z_0} f(z)=\infty$ Pol

$\lim_{z\to z_0} f(z)$ 不存在 wesentlich Singularität.

Res.

$res(f,z_0)=a_{-1}$

$res(f,z_0)=\lim_{z\to z_0}(z-z_0)f(z)$ 1阶极点.

$res(f,z_0)=\frac{1}{(n-1)!}\lim_{z\to z_0}\left((z-z_0)^n f(z)\right)^{n-1}$ n阶极点.

$res\left(\frac{g}{h},z_0\right)=\frac{g(z_0)}{h'(z_0)}$ 1级极点.

复曲线积分.

$\int_c f(z)\,dz=\int_0^1 f(z(t))\,z'(t)\,dt$

$z(t)=x(t)+iy(t)$

im Kreisrand.

$\begin{pmatrix}x(t)\\y(t)\end{pmatrix}=\begin{pmatrix}r\cos(t)\\r\sin(t)\end{pmatrix}+\begin{pmatrix}x_0\\y_0\end{pmatrix}$

$z(t)=z_0+re^{it}$, $\quad z'(t)=ire^{it}$

$\ln(z)=\ln(|z|)+i\arg(z)$

$\oint_c f(z)\,dz=2\pi i\sum_{i=1}^{m} res(f,z_i)$

在C内的 z_i 不含边界.

$\int_{-\infty}^{\infty} f(x)\,dx \Rightarrow 2\pi i\sum res(f,z_i)$

上半平面的 z_i

$\int_0^{\infty} f(x)\,dx \overset{\text{gerade}}{=} \frac{1}{2}\int_{-\infty}^{+\infty} f(x)\,dx$

其他

$=-\sum_{k=1}^{m} res(f(z)\ln(z),z_k)$

$z_k\notin[0,+\infty)$

Fourier

$w<0$

$\int_{-\infty}^{\infty} f(t)e^{-iwt}dt=2\pi i\sum_{z_k\,Im(z_k)>0} res(f(z)e^{-iwz},z_k)$

$w>0$

$\int_{-\infty}^{\infty} f(t)e^{-iwt}dt=-2\pi i\sum_{z_k\,Im(z_k)>0} res(f(z)e^{-iwz},z_k)$

$I=\int_0^{2\pi} R(\cos t,\sin t)\,dt$.

$f(z)=z^{-1}R\left(\frac{z+z^{-1}}{2},\frac{z-z^{-1}}{2}\right)$

$I=2\pi\sum_{k=1}^{m} res(f,z_k)$

$z_k\in B_1(0)$

拉普拉斯变换表.

$1\longrightarrow\frac{1}{z}$

$t\longrightarrow\frac{1}{z^2}$

$\sin wt\longrightarrow\frac{w}{z^2+w^2}$

$\cos wt\longrightarrow\frac{z}{z^2+w^2}$

傅立叶变换表.

$\chi_{[-h,h]}\longrightarrow 2h\,\mathrm{sinc}(hw)=2h\cdot\frac{\sin(hw)}{hw}$

$e^{-|t|}\longrightarrow\frac{2}{1+w^2}$